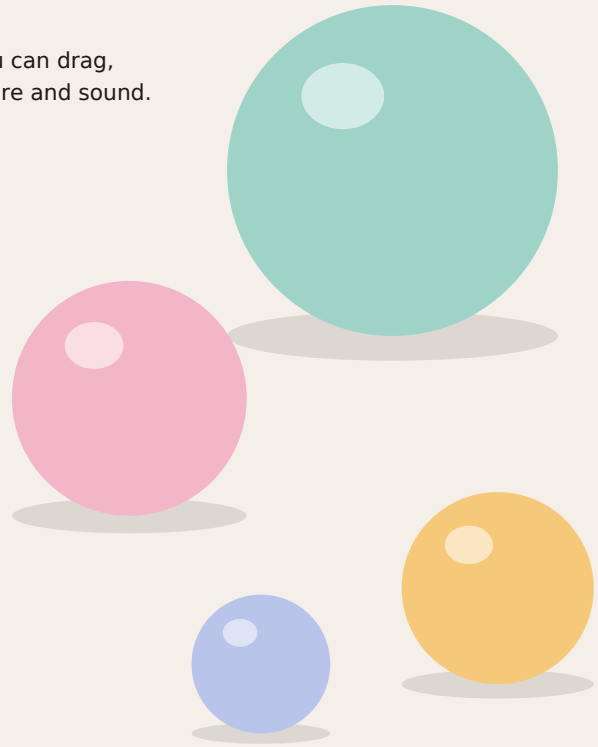
PROJECT BLUEPRINT / NO. 001

Wobble Studio.

Soft things, made playable.

A 3D website of 20 jelly, fruit and food objects you can drag, poke, squish and cut, each with its own feel, texture and sound.



OWNER
Rashedul Islam Nissan

BUILT WITH
Claude Code

HOSTING
Free static host, $0

01 / CONCEPT

What we are building

Wobble Studio is a playable 3D website. Soft objects sit on a quiet, tinted stage and respond to your hands and tools: they wobble, stretch, squash, crumble, snap and split. Each object belongs to a **material** that decides its feel, texture, sound and how it breaks. The tone is calm, tactile and editorial, inspired by the two reference videos (a jelly dolphin and a cuttable citrus slice), but with our own wording, names and visuals.

DECISION	LOCKED CHOICE
Site name	Wobble Studio
Objects	20 (9 jelly, 4 fruit, 5 soft foods, 2 firm foods). The "Puff" is an original character, not a franchise mascot.
Selector	Along the **top** of the page, grouped by material, 20 thumbnails.
Tools and controls	Floating panel on the **right** ("The Specimen"), contents change per material.
Background	**Tint changes with each object**, chosen so the object, its shadow and edges stand out clearly.
Physics	Custom XPBD soft-body solver: GPU compute on WebGPU, CPU fallback; WebGL2 rendering fallback.
Stack	Vite + TypeScript + Three.js, plain DOM/CSS UI, Web Audio, self-hosted fonts.
Budget	$0: static hosting only, CC0 assets, no backend.

02 / PRINCIPLES

Design principles

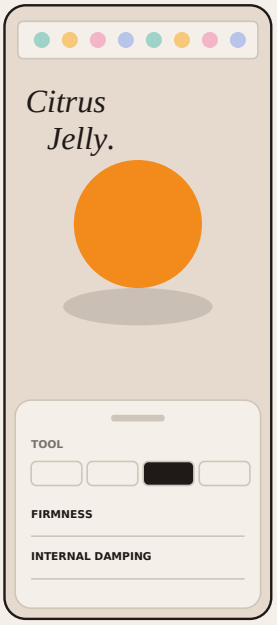
- **Feel first.** One blob that feels great beats twenty that feel dead. Tune damping, stiffness, drag response and sound before adding objects.
- **Materials are data.** New objects should mostly be a registry entry (mesh, material, tint, variants), not new engine code.
- **One object at a time.** Lazy-load, dispose on switch. This keeps phones smooth.
- **Always readable.** Every tint, shadow and label must be easy to see. Pale objects never vanish on pale backgrounds.
- **License-clean.** CC0 or clearly commercial-safe assets only, every one logged in CREDITS.md. No franchise characters.

03 / INTERFACE

Layout



DESKTOP



PHONE: slide-up sheet

AREA	WHAT GOES THERE
Top bar	Wordmark at left, scrollable strip of 20 thumbnails grouped Jelly / Fruit / Soft / Firm, sound toggle at far right. Arrow keys move between items.
Title block	Small tracked label, huge italic serif object name with an indented second line, one-line description.
Right panel	"The Specimen": Tool, Colour or Variety, Firmness and Internal damping sliders (mono value readouts), object actions, Nudge, Reset, 1/4 speed, Show mesh, Pause.
Bottom-left	Italic tool hint, then a live stats row: Mass, Volume (% of rest), Kinetic energy, Pieces. Tiny footnote below.
Status badge	Renderer in use (WebGPU or WebGL) and fps, small monospace.
Typography	Italic serif for titles and captions, tracked small-caps sans for labels, monospace for numbers. Hairline borders, sharp 2-6 px radii.

04 / OBJECTS

The 20 objects

#	OBJECT	GROUP	MATERIAL	WHERE IT COMES FROM	VARIANTS	SPECIAL ACTION
1	**Dolphin**	Jelly	jelly	Import CC0 GLB or generate	Atlantic, Lagoon, Rose	Kick flukes, Blow
2	**Gummy bear**	Jelly	jelly	Generate (rounded shapes)	Red, Green, Orange, Yellow	Bounce
3	**Octopus**	Jelly	jelly	Import CC0 GLB or generate	Plum, Coral, Mint	Wiggle arms
4	**Duck**	Jelly	jelly	Import CC0 GLB or generate	Lemon, Peach	Squeak
5	**Star**	Jelly	jelly	Generate (extrude + bevel)	Gold, Sky, Berry	Spin
6	**Heart**	Jelly	jelly	Generate (extrude + bevel)	Rose, Ruby, Blush	Heartbeat pulse
7	**Axolotl**	Jelly	jelly	Generate (body, tail, 6 gills)	Pink, Mint, Gold	Wiggle gills
8	**Penguin**	Jelly	jelly	Import CC0 GLB or generate	Classic, Ice, Sunset	Waddle
9	**Puffy (original)**	Jelly	jelly	Generate	Cotton pink, Lavender, Peach	Puff up
10	**Orange slice**	Fruit	fruit-juicy	Generate (wedges + rind)	Orange, Blood orange, Lemon	Cutter stamps
11	**Apple**	Fruit	fruit-crisp	Kenney Food Kit or generate	Red, Green, Gold	Bite crunch
12	**Watermelon**	Fruit	fruit-juicy	Generate (rind, flesh, seeds)	Classic, Yellow	Seed pop
13	**Tomato**	Fruit	fruit-juicy	Import or generate	Red, Yellow	Burst on hard press
14	**Cake**	Soft	sponge	Kenney Food Kit or generate	Vanilla, Chocolate, Strawberry	Crumble
15	**Bread**	Soft	sponge	Kenney Food Kit or generate	White, Wheat	Squish and spring back
16	**Donut**	Soft	sponge	Generate (torus + glaze)	Pink, Chocolate, Vanilla	Sprinkle burst
17	**Pudding**	Soft	cream	Generate (lathe + caramel)	Caramel, Chocolate, Matcha	Jiggle
18	**Mochi**	Soft	mochi	Generate (blob + tiny face)	White, Pink, Green	Stretch
19	**Cheese block**	Firm	firm-dairy	Import or generate	Swiss, Cheddar	Slice
20	**Chocolate bar**	Firm	brittle	Generate (segments)	Milk, Dark, Ruby	Snap

Puffy is an original round pink puffball: soft marshmallow-jelly body, a small curly tuft, huge sparkly glossy eyes, blush cheeks, a tiny smile and stubby feet. It puffs up on demand and makes a tiny squeaky sound when poked. It deliberately avoids any resemblance to existing franchise characters.

05 / FEEL

Materials

MATERIAL	FEEL	STIFFNESS	CUT BEHAVIOUR	PARTICLES	SOUND
jelly	Very wobbly, translucent, glossy	Low	Clean, soft, rippled faces	Shimmer	Squish, boing, wet slice
fruit-crisp	Firm, slight give, snappy	High	Clean, crisp resistance	Juice mist, chips	Crunch, snap
fruit-juicy	Plump, taut skin, juicy	Mid	Clean; tomato bursts	Droplets, pulp	Squelch, splat
sponge	Soft, springy, crumbly	Low-mid	Crumbles at edges	Crumbs	Soft thud, crumble
cream	Jiggly, smooth, creamy	Low	Smooth, closes slowly	Smear	Wobble-pluck, scoop
mochi	Stretchy, sticky, squashy	Low	Stretches before parting	Powder puff	Squeak, stretch creak
firm-dairy	Firm, smooth, slight spring	High	Very clean slice	Shavings	Dull thunk, smooth slice
brittle	Hard, almost no wobble	Very high	Snaps along segments	Shards, dust	Sharp snap, crack

06 / INTERACTION

Tools

TOOL	JELLY	FRUIT	SOFT / CREAM / MOCHI	FIRM
Hand (drag, toss)	✓	✓	✓	✓
Poke (finger)	✓	✓	✓	✓
Press (squish plate)	✓	✓	✓	✓
Knife	✓	✓	✓	✓
Wavy knife	✓	✓	✓	–
Cutter (star, heart, flower, round)	✓	orange, watermelon	cake, bread	–
Fork (pierce)	–	✓	✓	cheese
Spoon (scoop)	–	watermelon	pudding, cake, mochi	–
Hammer	–	apple, tomato, watermelon	–	chocolate

Cutting is built in three levels. **Level 1** ships first: objects are made of pre-split chunks joined by breakable links, so a knife pass separates them convincingly and cheaply. **Level 2** breaks solver constraints along the tool's swept surface and re-caps the cut faces with inner flesh or sponge texture. **Level 3** (optional) is true arbitrary mesh slicing. Chocolate never cuts: it snaps along its segment lines.

07 / VISIBILITY

Background tint system

Each object's background is the **complementary hue** of its main colour, with lightness chosen **against** the object so nothing disappears. Shadows are a darker, richer version of the background hue (never plain grey). A soft halo and rim light separate the edges, and a contrast guard auto-adjusts any tint that is too close. Pale objects (cake, mochi, cheese) get a mid-light, more saturated background. The swatches below are **illustrative starting points**; the app computes the final tints and the owner can hand-tune any of them via a per-object override. Each card shows background tint, spotlight, tinted shadow and the object.



Backgrounds crossfade over about 0.6 seconds when the object or variant changes. A dev page (`?dev=tints`) shows all 20 objects and variants side by side for review.

08 / ASSETS

Where Claude Code gets everything

TYPE	SOURCES	RULES
Models (import)	Kenney Food Kit (CC0), Quaternius (CC0), Poly Pizza (check each label), Sketchfab as last resort	Check every license. Decimate anything over about 3,000 triangles. Preprocess with a script in /scripts.
Models (generate)	Code: lathe, extrude, SDF / metaballs with marching cubes	If no good CC0 model is found quickly, generate instead of settling for a bad import.
Textures	ambientCG and Poly Haven (both CC0)	Apple skin, crumb, sponge, cheese, chocolate. Compress; 1024 px max unless it clearly matters. Jelly needs none.
Sounds	Web Audio synthesis (squish, boing, squeak, wobble); Kenney audio packs and Freesound CC0 filter (crunch, thud, crumble, snap, splash)	Pixabay SFX are free for commercial use but not CC0: log and verify the license text. Keep initial audio under about 500 KB.
Fonts	@fontsource packages, self-hosted	Italic serif, tracked sans, monospace. No CDN fonts.
Logging	CREDITS.md	Every external asset: name, author, source URL, license. Final licensing audit before launch.

09 / BUDGETS

Performance targets

- **60 fps** on desktop, **30 fps or better** on a mid-range phone, with an auto-quality governor that lowers solver iterations, particles, shadow resolution and render scale.
- One object loaded at a time; models, textures and audio load lazily; GPU resources are disposed on switch.
- Initial JS and CSS under about 400 KB gzipped (excluding the Three.js chunk); first object interactive in about 3 seconds.
- WebGPU when available with WebGL2 fallback; a friendly static page if neither exists.
- Audio starts muted until the first user gesture; the sound toggle remembers its state.

10 / PHASES

Build roadmap

PHASE	WHAT HAPPENS	DONE WHEN
0 Setup	Vite + TS repo, folders, fonts, CREDITS.md, README, deploy script.	Empty styled stage runs; build passes.
1 Physics spike	Hidden jelly blob (?dev=blob), XPBD solver, drag, toss, GPU and CPU paths. Decide tetra mesh vs lattice + skin.	Feels like real jelly, stable, 60 fps desktop and 30+ fps phone.
2 Stage and UI shell	Lighting, floor glow, tinted shadow, jelly material, tint system, top bar, title block, right panel, stats.	Blob on a tinted stage; UI matches the layout on desktop and phone.
3 Materials, audio, basic tools	Material system, audio engine, Hand/Poke/Press/Nudge. Dolphin, Orange, Apple, Cake, Pudding.	Each of the five is distinguishable by feel and sound alone.
4 Cutting	Knife, Wavy knife, Cutter stamps, Fork, Spoon, Hammer; pieces, particles, counters.	Five objects cut cleanly without physics blow-ups.
5 Remaining objects	Add the other 15 via the registry (mostly data).	All 20 load, play and match their materials.
6 Polish and launch	Loading states, bottom sheet, keyboard and screen reader, reduced motion, auto-quality, share, meta tags, fallback page, license audit, deploy.	Full checklist in CLAUDE.md passes.

11 / RISKS

Risks and how we handle them

RISK	WHY IT MATTERS	PLAN
Cutting plus soft bodies	Hardest part of the project; easy to destabilise.	Ship Level 1 first; only move up when stable. Cap pieces (about 24) and fade tiny shards.
Phone performance	Soft-body sims are heavy.	One object at a time, auto-quality governor, CPU fallback at reduced resolution.
WebGPU support	Many phones and browsers lack it.	WebGL2 rendering fallback and a CPU solver path from the start.
Asset licensing	Free is not always commercial-safe.	CC0 first, log everything, final audit, generate in code when unsure.
Tint readability	Translucent and pale objects can vanish.	Lightness chosen against the object, contrast guard, ?dev=tints review page.

HOW TO USE THIS WITH CLAUDE CODE

Put `CLAUDE.md` in the project root, open the folder in Claude Code, and start with: "*Read CLAUDE.md and the blueprint, then plan and build Phase 0 and Phase 1. Stop when the Phase 1 acceptance criteria pass and show me.*"